Report

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Introduction

This report is about the final project of the LEAP course for for 'Machine Learning and applications'. The project theme selected is the “Option 1”(build a network that recognizes handwritten digits, then study where it makes its mistakes).

Aim

The aim of this project is to build a model that can predict the digit from images of handwriting digits. To do so a convolutional neural network is utilized. Based on the knowledge gained though out this course, we have to train and evaluate the model and conclude on how a change in the model train or architecture can influence the results or predictions.

Post processing

Data

The data used for this project are from the MNIST dataset which is accessed thought the OpenML(openml.org). The dataset consists of 70000 images flattened to into one dimensional vectors of 784 numbers. We reshape them to 28x28 vectors with greyscale channel, meaning one channel. The values of the data are normalized by dividing by 255.

We split the data to ‘train’, ‘validation’ and ‘test’ set of [72,8,20] percentages.

The labels are provided as text which we convert to numbers to be easy to work with.

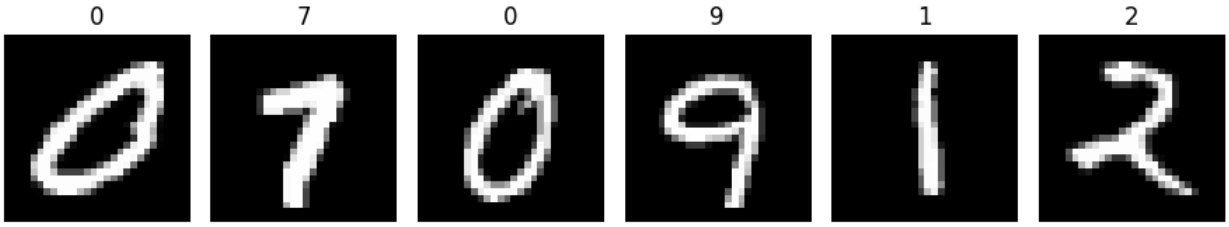


Figure 1: A snapshot of the dataset. Example of hadnwriten digits with their labels above

Training

Model architecture

For this task of image classification we use a CNN of the following architecture:

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 28, 28, 32)	320
max_pooling2d_3 (MaxPooling2D)	(None, 14, 14, 32)	0
dropout_3 (Dropout)	(None, 14, 14, 32)	0
flatten_3 (Flatten)	(None, 6272)	0
dense_6 (Dense)	(None, 64)	401,472
dense_7 (Dense)	(None, 10)	650

Figure 2: The architecture of the model.

Optimizer

The optimizer used is Adam with learning rate of 1e-3, loss function of sparse categorial cross entropy. For monitoring we are using the accuracy metric.

Early Stopping

We utilized and early stopping criterion to ensure that the model won't train for longer than necessary. The callback created for this cause stops the training when no improvement in the values of validation loss is detected for 5 consecutive epochs.

Evaluation of the trained model

We present the learning curves of a single training session. For more accurate and fair study of the model would be better to do multiple sessions and present the mean and standard deviation of these curves. The final accuracy is 0.9871.

The test accuracy is evaluated by using the data of the 'test' set for predictions. It is measured at 0.984

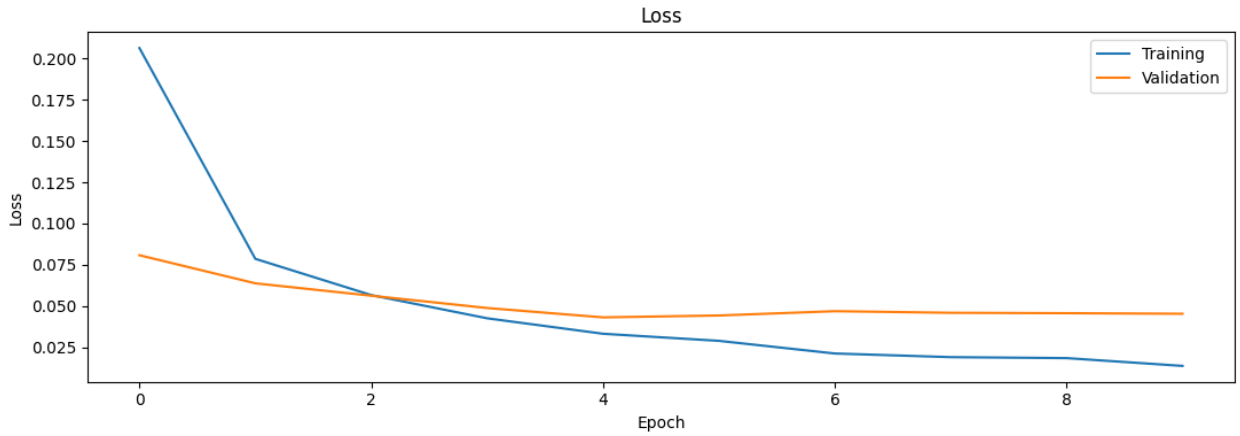


Figure 3 Learning curves of one training session

Figure 4 presents the confusion matrix. The matrix help as understand the weak points of the model by showing us which digits are more difficult for the model to predict correctly.

Some key findings from the matrix are:

- The digit zero is learned almost perfectly as only one instant is predicted wrong, misclassifying it as one.
- The digit four is the most misclassified, mainly as nine, which occurred 17 times.
- Digit 3 was misclassified as five, 11 times and was also classified as other digits giving a variant of predictions

The diagonal is strong as displayed by the high values of correct predictions which is encouraging and shows the success of the training.

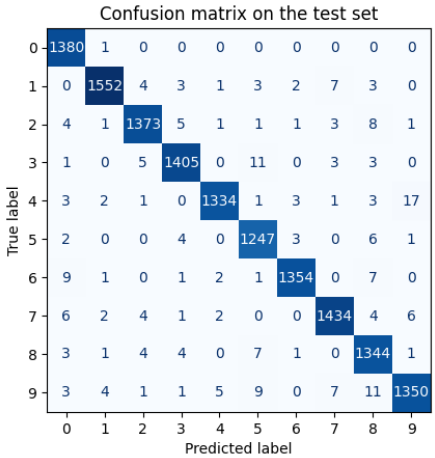


Figure 4 Confusion matrix

Study of changed model

We select the option B for the part of studying the influence of a change in the architecture to the performance of the model. We change drop rate from 0.3 to 0.5 and we retrain the model. We except to see a model is less prone to overfitting.

Figure 5 shows the confusion matrix of the new model. After training the final training accuracy is 0.971, slightly less than the initial model. The increase of dropout increased also the underfitting and removed representation capacity by forcing the model to rely on fewer neurons.

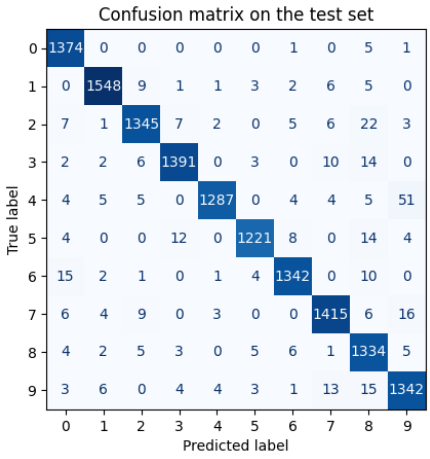


Figure 5 Confusion matrix after the change on the model

Conclusions

The first model makes the most of the mistakes for the digits four and three. By increasing dropout rate the model becomes prone to underfitting, letting fewer neurons gather the geometrical information of the digits. The new model predicted correctly 14 times less than the initial model the digit 3, thought it manage to reduce the confusion with digit 5 it increased the confusions with other digits. For digit 4 the correct predictions dropped by 47 and the confusion with digit 9 was even stronger by an increase of 34. This one change can't be taken into consideration as it was only one training session and can't guaranty meaningful conclusions. For better validation it should be tested multiple times to compare the mean and standard deviation of the results.